Badr ALKHAMISSI
PhD Candidate at EPFL

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https://bkhmsi.github.io github.com/bkhmsi linkedin.com/in/bkhmsi scholar.google.com/bkhmsi

I am passionate about building intelligent systems that can solve complex machine learning tasks and at the same time reveal insights about the inner workings of biological agents. Such systems do not only pave a way for solving long-standing societal problems but also shed light on what it means to be human.

Please see website for a more detailed resumé.

Education

Present – September 2023 | **EPFL, PhD, Lausanne, Switzerland**
- Co-advised by Antoine Bosselut (NLP Lab) and Martin Schrimpf (NeuroAI Lab).

September 2020 – September 2019 | **Goldsmiths, University of London, Master of Science, London, UK**
- MSc. in Computational Cognitive Neuroscience - Pass with Distinction.
- Thesis: Episodic Meta-Reinforcement Learning on the Two-Step and Harlow Tasks (Grade: 90%).

June 2018 – September 2014 | **The American University in Cairo, Bachelor of Science, Cairo, Egypt**
- BSc. in Computer Science - Summa Cum Laude. **GPA 4.0**
- Thesis: Video Search via Natural Language.

Selected Research and Industrial Experience

Present – July 2026 | **Student Researcher, Google DeepMind, Zurich, Switzerland**
- Currently a student researcher at Google DeepMind.

Machine Learning | Latent Reasoning | Recursive Transformers | AI for Science

November 2022 – November 2021 | **AI Resident, Meta AI, Seattle, US**
- Research in NLP. Published 5 first-author papers, and 1 co-author paper. (See Publications)

NLP | Hate Speech | Responsible AI | Reasoning | Arabic NLP | FAIR

October 2021 – October 2020 | **Research Intern, Sony AI, Tokyo, Japan (Remotely)**
- Published 2 papers in collaboration with DeepMind and 1 paper with Sony CSL. (See Publications)

Meta-learning | Reinforcement Learning | Symbolic Alchemy

September 2019 – April 2019 | **Applied Scientist, Microsoft Advanced Technology Lab in Cairo, Cairo, Egypt**
- Part of the LUIS Applied Science Team and the Machine Teaching Innovation Group.

NLU | NLP | MT | Keras | Tensorflow | PyTorch | Python | C#

Selected Publications

Selected Conferences

Discovering Functionally Selective Brain Regions with a Deep Topographic Multimodal Model — CCN 2026
Spotlight Paper Website github.com/https://github.com/epflneuroailab/topo-omni Tweets
Authors: **Badr AlKhamissi***, Johannes Mehrer*, Lara Marinov, Ahmed Abdelaal, Abdulkadir Gokce, Martin Schrimpf

Large Language Models Align with the Human Brain during Creative Thinking — COLM 2026
Paper
Authors: Mete Ismayilzada, Simone Luchini, Abdulkadir Gokce, **Badr AlKhamissi**, Antoine Bosselut, Antonio Laverghetta Jr, Lonneke van der Plas, Roger Beaty

Apertus: Democratizing Open and Compliant LLMs For Global Language Environments — ACL 2026
Paper Media Models Tweets
Authors: Alejandro Hernández-Cano, Alexander Hägele, [...], **Badr AlKhamissi**, [...], Antoine Bosselut*, Imanol Schlag*, Martin Jaggi*

Mixture of Cognitive Reasoners: Modular Reasoning with Brain-Like Specialization — ICLR 2026
Paper Website Tweets github.com/https://github.com/bkhmsi/mixture-of-cognitive-reasoners
Authors: **Badr AlKhamissi**, C. Nicolò De Sabbata, Zeming Chen, Martin Schrimpf*, Antoine Bosselut*

Inducing Dyslexia in Vision Language Models — ICLR 2026
Paper Tweets github.com/https://github.com/epflneuroailab/VWFA-Localization
Authors: Melika Honarmand, Ayati Sharma, **Badr AlKhamissi**, Johannes Mehrer*, Martin Schrimpf*

From Language to Cognition: How LLMs Outgrow the Human Language Network EMNLP 2025
Oral Paper Tweets
Authors: **Badr AlKhamissi**, Greta Tuckute, Yingtian Tang, Taha Binhuraib, Antoine Bosselut*, Martin Schrimpf*

The LLM Language Network: A Neuroscientific Approach for Identifying Causally Important Units NAACL 2025
Oral Paper Website github.com/https://github.com/bkhmsi/llm-localization
Authors: **Badr AlKhamissi**, Greta Tuckute, Antoine Bosselut*, Martin Schrimpf*

TopoLM: Brain-Like Spatio-functional Organization in a Topographic Language Model ICLR 2025
Oral Paper Tweets
Authors: Neil Rathi*, Johannes Mehrer*, **Badr AlKhamissi**, Taha Binhuraib, Nicholas M. Blauch, Martin Schrimpf

Investigating Cultural Alignment of Large Language Models ACL 2024
Paper Tweets github.com/https://github.com/bkhmsi/cultural-trends
Authors: **Badr AlKhamissi**, Muhammad ElNokrashy, Mai AlKhamissi, Mona Diab

A Context-Contrastive Inference Approach To Partial Diacritization ArabicNLP 2024
Oral Paper Demo github.com/https://github.com/munael/ccpd-partial-arabic-diacritization
Authors: Muhammad ElNokrashy, **Badr AlKhamissi**

Instruction-tuning Aligns LLMs to the Human Brain COLM 2024
Paper Tweets
Authors: Khai Loong Aw, Syrielle Montariol*, **Badr AlKhamissi***, Martin Schrimpf, Antoine Bosselut

Depth-Wise Attention (DWAtt): A Layer Fusion Method for Data-Efficient Classification LREC-COLING 2024
Oral Paper Poster
Authors: Muhammad ElNokrashy*, **Badr AlKhamissi***, Mona Diab

"Flex Tape Can't Fix That": Bias and Misinformation in Edited Language Models EMNLP 2024
Paper
Authors: Karina Halevy, Anna Sotnikova, **Badr AlKhamissi**, Syrielle Montariol, Antoine Bosselut

ALERT: Adapt Language Models to Reasoning Tasks ACL 2023
Paper Poster
Authors: Ping Yu, Tianlu Wang, Olga Golovneva, **Badr AlKhamissi**, Siddharth Verma, Zhijing Jin, Gargi Ghosh, Mona Diab, Asli Celikyilmaz

ToKen: Task Decomposition and Knowledge Infusion for Few-Shot Hate Speech Detection EMNLP 2022
Oral Paper Tweets
Authors: **Badr AlKhamissi***, Faisal Ladhak*, Srini Iyer, Ves Stoyanov, Zornitsa Kozareva, Xian Li, Pascale Fung, Lambert Mathias, Asli Celikyilmaz, Mona Diab

Selected Workshops

A Self-Synthesis Approach For Training Vision-Language Models With Developmentally Plausible Data BabyLM 2024
CoNLL 2024 Paper
Authors: **Badr AlKhamissi***, Yingtian Tang*, Abdülkadir Gökce*, Johannes Mehrer, Martin Schrimpf

Khattat: Enhancing Readability and Concept Representation of Semantic Typography AI4VA 2024
ECCV 2024 Paper Website Tweets
Authors: Ahmed Hussein*, Alaa Elsetohy*, Sama Hadhoud*, Tameem Bakr*, Yasser Rohaim*, **Badr AlKhamissi**

OPT-R: Exploring the Role of Explanations in Finetuning and Prompting for Reasoning Skills of LLMs NLRSE 2023
ACL 2023 Paper Poster Tweets
Authors: **Badr AlKhamissi**, Siddharth Verma, Ping Yu, Zhijing Jin, Asli Celikyilmaz, Mona Diab

MultiTask Learning with Self-Correction for Hate Speech Classification OSACT5 2022
LREC 2022 Oral Paper Best Paper Award Tweets
Authors: **Badr AlKhamissi***, Mona Diab

How to Learn and Represent Abstractions: An Investigation using Symbolic Alchemy AutoML Workshop 2022
AutoML 2022 Paper Poster github.com/https://github.com/BKHMSI/alchemist
Authors: **Badr AlKhamissi**, Akshay Srinivasan, Zeb Kurth-Nelson, Sam Ritter

The Emergence of Abstract and Episodic Neurons in Episodic Meta-RL MemARI 2022
NeurIPS 2022 Spotlight Oral Paper Poster
Authors: **Badr AlKhamissi**, Muhammad ElNokrashy, Michael Spranger

Selected Preprints

MIRAGE: Adaptive Multimodal Gating for Whole-Brain fMRI Encoding Under Review
Paper Website Tweets github.com/https://github.com/epflneuroailab/mirage
Authors: Abdulkadir Gokce*, **Badr AlKhamissi***, Martin Schrimpf

MEDITRON: Open Medical Foundation Models Adapted for Clinical Practice Preprint 2024

Paper github.com/https://github.com/epfLLM/meditron

Authors: Zeming Chen, Angelika Romanou*, Antoine Bonnet*, Alejandro Hernández-Cano*, **Badr AlKhamissi***, Kyle Matoba*, [...], Physician Evaluation Group, Noémie Boillat-Blanco, Kristina Keitel, Javier Elkin, Blaise Robert, Syrielle Montariol, Mary-Anne Hartley*, Martin Jaggi*, Antoine Bosselut*

A Review of Language Models as Knowledge Bases Preprint 2022

Paper Website Tweets

Authors: **Badr AlKhamissi***, Millicent Li*, Asli Celikyilmaz, Mona Diab, Marjan Ghazvininejad

Selected Awards

- **Google PhD Fellowship** in NLP Recipient. '26
- **KAUST Rising Stars in AI** Recipient. '26
- **Swiss AI Initiative PhD Fellowship** Recipient. '25
- **EPFL Doc.Mobility Grant** Recipient. '25
- **President's Cup** and **Mohamed El Beleidy Academic Award** for class valedictorian - American University in Cairo '18.
- **Mohammed Bin Abdulkarim A. Allehedan Award** for highest performing STEM student - American University in Cairo '18.
- **First place** on the two categories of the Arabic Reverse Dictionary shared task at ArabicNLP 2023. '23
- **Best Paper Award** for the Fine-Grained Hate Speech Detection shared task in Arabic. '22
- **First place** in all four categories of the second NADI shared task. '21
- **Best Mentor Award** at Microsoft Advanced Technology Lab in Cairo. '19
- **Computer Science Exemplary Student** - American University in Cairo '18
- **Outstanding Performance Award** in Deep Learning Summer School at Tsinghua University. '17

Teaching Experience

Spring '24,'25,'26 & Fall '24 — **Teaching Assistant, EPFL, Lausanne, Switzerland**

- Teaching assistant for Modern Natural Language Processing and Linear Algebra.

Python | PyTorch | NLP | LLMs | RLHF

Fall '15 & '17 & Spring '16 — **Undergraduate Teaching Assistant, The American University in Cairo, Cairo, Egypt**

- Teaching assistant for Deep Learning and Programming Fundamentals courses.

C++ | Python | Tensorflow | Keras | SFML

Selected Invited Talks

- **IndabaX Sudan Featured Speaker**: Building Responsible & Ethical LLMs.
- **LauzHack Deep Learning Bootcamp**: Brain-Inspired Large Language Models.
- **REST-CL Retreat for Students in Computational Linguistics (UPF University)** The LLM Language Network.
- **Psychology Days at Riga Stradins University**: Vision and Language in Brains, Minds and Machines.
- **Nancy Kanwisher's Lab at MIT**: Mixture of Cognitive Reasoners.
- **NCCR Evolving Language Group**: Brain Alignment of Untrained Models.
- **Cohere For AI Community Talk (Geo Regional Africa Group)**: Investigating Cultural Alignment of LLMs.
- **ARBML Board Talk**: Investigating Cultural Alignment of LLMs.
- **AiBelmasry Podcast** (Guest Speaker): Analyzing LLM's Ability in Reasoning.
- **TEDxMIU Talk**: Representation Matters: Celebrating Egyptians in the AI Revolution.
- **Cohere For AI Community Talk**: Presenting OPT-R Paper.
- **Mada Masr Podcast Guest Speaker (April 6th 2023 Episode)**: A New World Shaped by Artificial Stupidity.
- **IndabaX Egypt 2019** Jointly led a tutorial on cross-lingual alignment.

Service

Organizing Committee

ICLR 2026 Workshop on Representational Alignment (Re-Align). ArabicNLP 2025 (Publicity Chair). ICML Workshop on Large Languages and Cognition (2024). ArabicNLP 2024 (Social Chair).

Journal Reviewer & Selected Program Committee Member (Reviewer)

Nature Communications. Neurobiology of Language. Nature Machine Intelligence. TACL. OpenMind. EMNLP 2026 (Area Chair). CogSci 2026. EACL 2025. NeurIPS 2025. EMNLP 2025. ACL 2025. NAACL 2025 (Area Chair). EMNLP 2024. NeurIPS 2024. ArabicNLP 2024. CCN 2024. ACL 2024. NAACL 2024. EMNLP 2023. ACL 2023. ENLSP-II Workshop at NeurIPS 2022. EMNLP 2022. AutoML 2022. ICBINB at NeurIPS 2021. Meta-Learn at NeurIPS 2021, 2020. L2L at ICLR 2021.